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Striking the balance between fake and real: under what conditions can media literacy messages that warn about misinformation maintain trust in accurate information?

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ABSTRACT

Although interventions that warn about the harms of misinformation may be effective in lowering the credibility of false information, they may also cause suspicion related to factually accurate information. To explore these contradictory outcomes of exposure to media literacy messages, we used an online survey experiment with a diverse sample of 1105 participants in the U.S. In this experiment, we randomly exposed people to traditional warning messages about the threats of misinformation or relativising warning messages that placed misinformation's threat in the context of the abundance of honest information. We additionally varied the specificity of the warning message (i.e. topic specific versus generic). We did not find direct overall effects on truth discernment but observed conditional effects on decreasing beliefs in misinformation and negative spillover effects on the truth rating of accurate information. We conclude that the effectiveness of media literacy interventions is far from straightforward, and document how pre-existing media trust plays a key role in the effects of such interventions. Based on our findings, we suggest that the effectiveness of specific and general media literacy messages may be contingent upon tailoring the message to levels of existing (dis)trust.

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Responding to the allegedly unprecedented threats of mis- and disinformation, different interventions to 'fight' false information have been developed and applied in recent years. Prominent examples include fact-check messages that refute falsehoods after their spread (e.g. Nyhan et al. 2020; Walter et al. 2020) and pre-bunking or inoculation messages that warn people about false messages before they are encountered (e.g. Hameleers 2022; Roozenbeek and van der Linden 2019). Although experimental studies show how both strategies are effective in correcting misperceptions and lowering the acceptance of false statements (see e.g. Walter et al. 2020), emphasising the threats of mis- and disinformation in interventions may also reduce trust in factually accurate information (Acerbi, Altay, and Mercier 2022; Modirrousta-Galian and Higham 2022).

The urgent problem we witness is that trust in the media and established information is low whereas concerns about the impact of misinformation are

alarmingly high (Newman et al. 2023). Given that risk perceptions related to the prevalence of misinformation are potentially not proportionate to the scope of false information encountered by media users (Knuuttila, Neudert, and Howard 2022), it is crucial to assess whether warning people about the risks of misinformation is spilling over to rising distrust in established information (Van der Meer, Hameleers, and Ohme 2023). Therefore, this paper investigates the effect of warning about misinformation on the credibility of both misinformation and accurate information. In addition, we explore whether an alternative warning strategy that relativises the threats of misinformation can prevent potentially negative side effects of warning messages.

Although the potential of warning messages to enhance distrust in factually accurate information has been demonstrated in extant research (e.g. Modirrousta-Galian and Higham 2022; Van der Meer, Hameleers, and Ohme 2023), we lack research on how this

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effect can be mitigated, and what forms of intervention strategies may be most effective in stimulating truth discernment. As a next step in the literature, we postulate that interventions should be designed in a way that they can help citizens to discern between factually accurate information and falsehoods, whilst consolidating trust in authentic and reliable sources of information. Here, in comparison to existing research on interventions (e.g. Van der Meer, Hameleers, and Ohme 2023), we look at the effects of interventions on the rating of both true and false information. Considering that less than one percent of all information reaching citizens in the digital media landscape may be considered as mis- or disinformation (Acerbi, Altay, and Mercier 2022; Allen et al. 2020), an emphasis on the omnipresence of ‘fake news’ and media manipulation may trigger suspicion related to all information encountered, and cultivate disproportionate levels of concerns among media users.

With this experiment, we aim to offer evidence-based suggestions on which types of media literacy interventions may be most effective in cultivating resilience towards mis- and disinformation. By looking at the impact of media literacy messages on the perceived accuracy and credibility of verified and falsified political statements, we are able to assess the extent to which interventions may simultaneously focus on the untrustworthiness of false information and the reliability of real information. In the remainder of this paper, we will first of all review the literature on misinformation and the effects of warning messages that contain a media literacy intervention. Moving beyond existing literature that has mapped the intended and unintended effects of such warning messages, we then propose an intervention that relativises the threats of misinformation. In addition, we suggest that a differentiation between issue specific and general warning messages may play a role in truth discernment. After this theoretical discussion, we discuss the set-up and findings on an experiment in which we explored the effects of different types of warning messages containing a traditional or relativising warning message, and a specific or general warning. We end with a discussion on the implications of our findings for media literacy interventions in an age of increased concerns on the global spread of misinformation.

1.1. Misinformation and its wider epistemic implications

In this paper, we define misinformation as an overarching term for information that is factually inaccurate, or not based on relevant expert knowledge and empirical

evidence (also see Vraga and Bode 2020). Generally, misinformation refers to information that lacks facticity, and this lack of facticity may be the result of different factors, such as a lack of expert consensus, the intention to influence elections or the intention to make a financial profit. Hence, this comprehensive conceptualisation may involve both unintentional untruthfulness, which has also been labelled as misinformation (Wardle 2017), and goal-directed deception, manipulation or untruths, typically referred to as disinformation (Freelon and Wells 2020). Although the distinction between misinformation and disinformation is conceptually relevant, the intentions of the communicator to deceive are difficult to establish, especially when the wider context or planned behaviours of malign actors are unclear (Chadwick and Stanier 2022; Hameleers 2023). Considering that this paper focuses on false information on a statement level, and therefore neglects the context of deception (i.e. the wider embedding of false information in election campaigns or its dissemination via hyper-partisan platforms), we use the term misinformation as an umbrella term that refers to untruthful information – which can be intentionally deceptive in certain contexts. Hence, based on inaccurate and non-factual statements alone, we cannot make inferences about the intentions or causes underlying the falsehoods communicated to people (Hameleers 2023). Thus, the false information shown to people in this study may be both misinformation and disinformation, depending on the context of deception.

Misinformation does not only exist as false information, or a genre of untruthfulness (Egelhofer and Lecheler 2019) but may also pertain to a wider epistemic disorder in which the reliability of truthful information and established knowledge is questioned or even intestinally delegitimised. This epistemic disorder may consist of more general concerns and wider debates about the alleged destabilising influence of the presence of false information (Van der Meer, Hameleers, and Ohme 2023) but also as a more active strategy using disinformation as a delegitimising label in contexts of low trust in the honesty of established information (Waisbord 2018).

First, regarding the more intentional use of the label of misinformation, in a communication setting where misinformation is said to be dangerous and omnipresent, different media outlets, experts, politicians, or other actors are constantly blamed for spreading misinformation or fake news (e.g. Egelhofer and Lecheler 2019; Farhall et al. 2019). The mere existence of the threat of misinformation may thus be strategically exploited to delegitimise inconsistent truth claims. This is not without consequences: Accusations of

misinformation and fake news are found to lower trust in factually accurate information (Egelhofer et al. 2022; Van der Meer, Hameleers, and Ohme 2023; Van Duyn and Collier 2019). Against this backdrop, the threats of misinformation are not restricted to false information but also pertain to an epistemic crisis related to the relative status of facts and the delegitimisation of truthful information.

Second, misinformation's impact on the reliability and credibility of factually accurate information is not always intentional and orchestrated by malicious actors. Hence, interventions that are designed with intentions to combat misinformation, such as media literacy messages, may have an unintended negative spillover effect as they potentially lower the credibility of factually accurate information (Acerbi, Altay, and Mercier 2022; Hameleers 2022; Modirrousta-Galian and Higham 2022). By warning people about the existence of misinformation, people may become disproportionately aware of and sensitive to deception. Consequentially, their heightened sense of being deceived by information may spill-over to their ratings of factually accurate news. Given that extant research indicates that factually accurate information is substantially more prevalent in people's news diets than misinformation (Acerbi, Altay, and Mercier 2022), such a spill-over effect is potentially problematic. An important challenge in this regard is to formulate interventions and warnings in such a way that people are equipped with the right competencies to correctly distinguish between information that is accurate and reliable versus information that is false. But how substantial is the spill-over effect of warning messages, and how can we improve them in order to consolidate trust in factually accurate information?

1.2. The effects of media literacy messages

Different approaches to fighting or preventing the effects of misinformation are found to be successful. Fact-checks that check dubious or suspicious claims after they are spread are found to lower misperceptions (e.g. Hameleers and Van der Meer 2020; Nyhan et al. 2020; Walter et al. 2020). However, it requires a lot of effort to create a fact-check of a single deceptive statement, whereas the deceptive statement it responds to yields more engagement than the correction. In addition, fact-checks are only received by a relatively small audience segment: People who are more inclined to support the issue positions of misinformation are less likely to select a fact-check message that is incongruent with their identities, values, or existing beliefs (Hameleers and Van der Meer 2020). Thus, although

fact-checks can be effective once selected, they may not reach individuals that are in need of a correction most.

An alternative to fact-checking are pre-bunking messages – which has either been studied in the realm of inoculation strategies (Roozenbeek and van der Linden 2019) or media literacy messages that warn people about misinformation (e.g. Vraga, Bode, and Tully 2022). Inoculation is typically applied to gamified approaches that expose people to a small dose of misinformation and suggestions on how to recognise falsehoods (Roozenbeek and van der Linden 2019). Media literacy messages are generally placed in the information ecology as public service announcements, or strategically located in information settings where misinformation is likely to be encountered, such as social media discourses in crisis times. The underlying assumption is that pre-bunking messages that are detached from individual misinformation narratives are effective as they equip news users with the skills to act as fact-checkers themselves (see e.g. Van der Linden 2023). Thus, the concrete suggestions offered and the increased knowledge about the causes, content and consequences of misinformation should instill resilience to misinformation on a societal level (also see Jones-Jang et al. 2021).

Experimental research has consistently shown that media literacy or inoculation messages are effective in lowering the credibility of misinformation (e.g. Hameleers 2022; Roozenbeek and van der Linden 2019; Vraga, Bode, and Tully 2022). However, just like fact-checks that debunk disinformation, pre-bunking interventions are not without potential drawbacks. Most important, although most studies have measured their effectiveness as lowering the acceptance of misinformation, general warnings may also have a significant effect on lowering the perceived credibility of factually accurate news (Hameleers 2022). In a similar vein, Modirrousta-Galian and Higham (2022) found that although gamified inoculation interventions can lower the credibility of misinformation, these interventions do not improve the discernment between false and true headlines. The findings of Guess et al. (2020) are more optimistic: They found that a short warning message did increase discernment between true and false news in India and the U.S. Against the background of these different conclusions, we can conclude that evidence on the effectiveness of media literacy messages is debatable at best. This begs the question on how media literacy messages can be improved to instill resilience towards misinformation whilst maintaining trust in reliable and factually accurate information.

Some of the contradictory findings of previous research may be explained if we look in detail how

media literacy messages and inoculation messages are designed. Most interventions strongly focus on deception and offer tips on how ‘you can spot fake news’ (Vraga, Bode, and Tully 2022). Although misinformation is a marginal phenomenon when viewed in the context of all information (Acerbi, Altay, and Mercier 2022), the emphasis on ‘fake’ and ‘deception’ in current interventions may potentially inflate the threats of misinformation and activate more cynical and distrusting beliefs than necessary for the discernment between truthful and unreliable information. Psychologically, we can approach this problem as the disproportionate activation of suspicion in the context of the truth-default-theory (TDT) (Levine 2014). The TDT postulates that, in a communication context where most information is reliable, citizens need shortcuts in order to navigate their overburdened information setting. Therefore, they may accept the honesty of information *unless* deception is actively triggered.

In that regard, media literacy messages that warn about deception, fake news and dishonesty may trigger suspicion and motivate people to become sensitive to deception – which may be used in response to both factually accurate (Van der Meer, Hameleers, and Ohme 2023) and deceptive information. Against this backdrop, we formulate the following hypotheses on the effects of media literacy messages that warn recipients about false information:

H1a: Exposure to a media literacy intervention that warns about misinformation lowers the credibility of misinformation.

H1b: Exposure to a media literacy intervention that warns about misinformation lowers the credibility of factually accurate information.

As a more overarching question, we are interested in the relative effect of media literacy interventions on enhancing the accuracy of people’s truth assessments. We thus aim to assess if the positive effect on lowering the credibility of misinformation is not cancelled out by negatively affecting trust in real information – especially when taking into account that factually accurate information outweighs misinformation in people’s media diets (Acerbi, Altay, and Mercier 2022). We specifically aim to assess if the positive implications of warning about misinformation can outweigh the negative ramifications, or whether the impact on factually accurate news makes these interventions ineffective. We ask:

RQ₁: Does exposure to media literacy interventions have a positive overall effect on participants’ ability to discern misinformation from factually accurate information?

1.3. Consolidating trust whilst warning about potential deception

To redesign existing approaches into more effective media literacy interventions, we argue that warning messages should make people critical to information, but not overtly cynical. This corresponds to the idea of media literate citizens that are reflexive of the production and consumption of information, and the set of skills and competences needed for citizens to thoughtfully evaluate the content, effects and causes of information (e.g. Livingstone 2014). In line with this, we propose that effective media literacy messages should maintain a balance between ensuring trust in reliable information and scepticism towards dubious claims, whilst relativising the threats of misinformation in a context where most information can be trusted. Thus, we propose that interventions should not only emphasise the threats and risks of false information, as typically done in existing interventions (Hameleers 2022) but also emphasise that misinformation exists in a context of predominately reliable and accurate information. Against this backdrop, for this experiment, we have designed media literacy messages that inform people about the relative salience of misinformation as compared to factually accurate information. Herewith, we aim to warn people about misinformation in context, and offer more information on the relative scope of the threat (Acerbi, Altay, and Mercier 2022). But why would such a relativised message be effective?

The truth-default-theory (Levine 2014) postulates that people are generally more likely to accept the trustworthiness of information rather than to doubt its honesty. This default state of accepting the honesty of information should correspond with the actual base rate of deception in people’s newsfeed. Hence, as 99% of all information may be honest (Acerbi, Altay, and Mercier 2022), the truth default should result in accurate assessments of the trustworthiness of information in most cases. However, the truth-default theory also postulates that people may deviate from the default state of accepting the honesty of information when a trigger for suspicion is offered (Levine 2014). Such a trigger of suspicion may be presented by a media literacy message that warns people about the threats of misinformation without placing the risks of false information in context (Van der Meer, Hameleers, and Ohme 2023).

We expect that a relativised media literacy message that warns about the likelihood of misinformation whilst emphasising that most information is reliable should result in a more proportional level of scepticism, and thus a more accurate discernment between true and false information. Hence, interventions that more closely resemble the media ecology where a small amount

of misinformation is presented alongside a majority of truthful content should not make people overly sensitive to deception. As indicated by Burgoon et al. (1994), the priming of suspicion can lower accuracy ratings, which implies that a strong emphasis on ‘fake news’ and deception in conventional interventions lowers accuracy. At the same time, media literacy interventions should contribute to the democratic ideals of a critical citizenry that can detect faulty information (e.g. Mason, Krutka, and Stoddard 2018). Therefore, we propose that media literacy interventions that (a) relativise the threat of misinformation; (b) offer suggestions on how to recognise trustworthy information whilst (c) emphasising the need to be a critical news user are more effective than interventions that merely warn about misinformation. We hypothesise:

H2: Media literacy interventions that relativise the threats of misinformation whilst promoting trustworthy news are more effective than conventional interventions that only warn about misinformation.

1.4. General versus specific warnings about misinformation

Most research that has looked at the effects of media literacy messages has looked at general warnings, emphasising the threats of misinformation and suggestions on how to detect it without specifying the issue or context of falsehoods (e.g. Guess et al. 2020; Hameleers 2022; Moore and Hancock 2022). In practice, however, warning labels are frequently used to warn people about misinformation in the context of certain prominent events, such as elections, COVID-19, or the war in Ukraine. Meta, for example, used specifically designed warning tags to emphasise the threats of misinformation related to COVID-19 (Ling, Gummadi, and Zannettou 2023; Solon 2020). Yet, to date, we know little about the relative effectiveness of general versus more specific warnings.

In this setting, we compare the effects of general media literacy interventions warning about misinformation as a wider threat and specific labels that focus on misinformation related to the war in Ukraine specifically – a highly salient topic associated with misinformation at the time of data collection (e.g. Torman et al. 2022). Arguably, general warnings should be more effective across the board, as people can apply them to any issue covered in the news. Specific warning messages should, in contrast, be most effective in enabling discernment between false and true statement on the topic they relate to, whereas they should be less effective than general warnings for non-related

issues. We therefore postulate the following hypotheses:

H3a: General warnings in media literacy interventions are more effective than specific warnings across different issues.

H3b: Specific warnings in media literacy interventions are more effective for truth discernment on related statements than general warnings.

2. Method

To test our hypotheses, we use a pre-registered online survey experiment. The anonymous link to the preregistration can be found here: https://aspredicted.org/63M_MB3. In the experiment, we randomly exposed participants to either a regular media literacy intervention warning about misinformation in general or a relativised media literacy intervention relativising the threats of misinformation. As a second experimental factor, we varied the specificity of the media literacy message: The interventions either referred to false/trustworthy information in general, or false/trustworthy information related to the war in Ukraine. In the control group, a warning message was absent. The design can hence be summarised as a 2 (Media literacy intervention: Emphasising the threats of misinformation versus a relativised warning) $\times$ 2 (Specificity of the media literacy intervention: Generic versus issue-specific) + control group (the absence of a media literacy intervention).

2.1. Participants

The research company Dynata was used to recruit a diverse sample of U.S. participants matching the composition of society as close as possible regarding age, gender, level of education and ideological preferences. Dynata works with voluntary opt-in samples, and panels are limited regarding the amount of surveys they can participate in during a given period. In addition, their behaviours related to low-quality responses (i.e. extremely fast response times, biases in answer patterns that indicate a lack of attention) are closely monitored. In total, we achieved 1105 valid responses (response rate 76%). Of all participants retained in the final analyses, 52% were female. The average age was 46.71 ($SD = 17.23$). Regarding education, 32% of the sample was lower educated, 43% had a moderate education level, and 25% were highly educated. The distributions in the final sample by and large reflect the U.S. population. Appendix C shows the demographics broken down per condition.

2.2. Independent variables

For the design of media literacy interventions, we based ourselves on existing interventions used by different platforms and independent organisations (i.e. Facebook, established journalistic formats) (also see e.g. Clayton et al. 2020; Guess et al. 2020). For the conventional media literacy intervention, we stayed as close as possible to messages that were used by platforms to warn about the influence of false information, whilst offering concrete suggestions on how to recognise false information (also see Hameleers 2022). This media literacy message was entitled ‘How to spot fake news and misleading information’ and offered suggestion in the form of ‘Be skeptical of headlines’ (see Appendix A for the stimuli).

We adjusted this intervention for the relativised media literacy intervention, which was entitled ‘How to recognize trustworthy and accurate information.’ Although this intervention referred to similar suggestions to detect misinformation as the conventional media literacy intervention, it relativised the threats of misinformation with statements such as ‘Although a very small proportion of the information we encounter is fake or misleading, most regular news can be trusted.’ Moreover, the suggestion was formulated as enabling news users to recognise reliable information, whilst being sensitive to deceptive information (see Appendix A).

2.3. Dependent variables: the rating of (Mis)information

After reading one of the media literacy messages, or after reading an unrelated message in the control condition entitled ‘Museum to fix Ripples in Rembrandt’s Night Watch’, participants rated 16 different headlines on their veracity (by scoring them according to the following categories: ‘this is true’, ‘this is false’ or ‘unsure’). Participants rated eight statements about the Russian invasion of Ukraine (four true and four false headlines) and eight statements about other salient topics at the time of data collection, including crime rates, immigration and COVID-19 (again four true and four false headlines). We decided to use statements instead of full news articles as this allowed for the presentation of a more diverse set of claims for each participant. Although we believe that the presentation of headlines and short statements can be reflexive of the current digital news environment and online newsfeeds, we should note that judgements about truthfulness may be more informed and accurate when people can read the entire news item that offers a context for the presented false

statements. That being said, the statements were carefully selected by consulting different independent fact-checking platforms and we made sure that the entire false claim was covered in the statements. Thus, more than being a decontextualised or general headline, the headlines used in the experiment contained the entire statement that was fact-checked. We made sure that the true and false headlines were clearly classified as true and false by at least two fact-checking organisations. We also included a balanced selection of items relating to different ideological perspectives, and finally varied the level of fact-based claims (i.e. some contained clearer numbers than other statements, reflecting the variety of disinformation in real life).

We should note that the selection of statements can be subject to biases, and that we could have selected other statements in different contexts (i.e. the timing of the study may have resulted in the selection of headlines that were prominent at the time of data collection). To minimise the influence of bias, we developed different inclusion and exclusion criteria: (1) the statements needed to be falsifiable; (2) the statements should not refer to specific recent/unfolding events but rather reflect positions or reflections on past events; (3) the statements had to deal with well-known politicised positions on the different issues central in politics (i.e. anti-immigration, crime rates, COVID-19). The selected statements aimed to reflect the politicised misinformation landscape as close as possible. Hence, we based the selection of claims on existing literature analysing mis- and disinformation across issues such as Covid-19 (Brennen, Simon, and Nielsen 2021) and immigration (Bennett and Livingston 2018). Appendix B includes an overview of all the headlines selected for this experiment.

Based on the headlines that were rated, three dependent variables were created. First, the perceived credibility of misinformation was calculated by taking the sum of the credibility scores of the eight items containing misinformation claims (-1 = false, 0 = don’t know, 1 = true). Higher scores thus refer to perceiving misinformation as more credible. Second, the credibility of factual information was measured by summing the credibility scores of the eight items containing factual claims (-1 = false, 0 = don’t know, 1 = true). Finally, an overall accuracy score was constructed by taking the total sum of the credibility rating of the factually accurate and misinformation statements after recoding the credibility of misinformation variable (as higher credibility ratings reflect less accurate responses, this variable was reverse coded to reflect accurate rating). For the composite measure, higher scores reflect more accurate truth

Table 1. Overview of statements and participants' truth rating in percentages.

Statements	True (%)	False (%)	Don't know (%)
NATO officers are entering the Kharkiv region to organise interaction between foreign and Ukrainian units (1)	25	22	53
Chinese military convoys have been entering Ukraine to support Russia in the war (2)	20	36	44
North Korea has taken military action against Russia (3)	14	54	32
Russian nuclear targets for Britain mapped amid rising tensions after Ukraine invasion (4)	24	26	50
The FBI spent more time investigating the Jan. 6 Capitol attack than 9/11 (5)	38	29	33
Monkeypox is a complete fake sham that's designed to force lockdowns, mandates and shots (6)	15	60	25
Violent crime rates increased in all but two Wisconsin cities between 2008 and 2017 (7)	31	16	53
Texas has had over 600,000 crimes committed by illegals since 2011, which includes 1,200 homicides (8)	29	23	48
Ukrainian President Volodymyr Zelensky has remained in the country since Russia invaded on Feb. 24 (9)	56	14	30
Several megachurches in the U.S. have raised funds to support Ukrainians in light of the Russian invasion (10)	46	12	42
The war in Ukraine has contributed to global shortages of certain products (11)	64	15	21
Individual foreign fighters originating from NATO member states are present on the battlefield in Ukraine (12)	36	19	45
In Virginia, Black people are eight times (8X) more likely than white people to die of gun homicide (13)	42	17	41
Crime in the U.S. is reduced to its lowest level in 27 years (14)	14	60	26
In 2020, more journalists were killed in Mexico than in any other country in the world (15)	23	23	54
The number of COVID-19 deaths recorded so far in 2021 has surpassed the total for 2020 (16)	35	29	36

ratings of respondents. Table 1 includes an overview of how the credibility of all statements was assessed. Here, it can be noted that many participants indicated that they did not know whether a statement was true or false, which indicates the difficulty of the task to discern factually accurate information from misinformation.

2.4. Control variables

We controlled for three factors that are found to strongly influence the rating of true and false information and the effects of mis- and disinformation more generally (e.g. Zimmermann and Kohring 2020): Media trust, measured with five items asking about whether media are fair in their covering unbiased, tell the whole story, are accurate in their coverage, and separate facts from opinions ($M = 3.62$, $SD = 1.77$); political ideology, measured as a left-right self-placement scale ranging from 1 (left) to 10 (right) ($M = 5.96$, $SD = 2.85$); and political interest, measured with the statement 'In general, how interested are you in political issues and public affairs?' (7-point Likert scale) ($M = 4.89$, $SD = 1.92$).

2.5. Procedures

Participants entered the survey link via the e-mail invitation of the research organisation. First, they completed the procedures for informed consent (the project received ethical approval under number [blinded for peer review]). Participants that agreed with the terms and conditions were forwarded to the pre-treatment survey, which included questions on age, gender, education, ideological preferences and media use variables. After the pre-treatment survey, they were forwarded to the experimental stimuli block (randomly allocated to the media literacy interventions

versus the control condition with a low-arousal and non-political message on art). On the next screen, we included an open-ended thought listing question to make people consider the message they just saw. This question additionally served as a delay before presenting the measures of the dependent variables. After writing up their thoughts in an unstructured manner, participants were presented with a list of 16 headlines (in a random order), presented as allegedly coming from a newsfeed (see Appendix B for the headlines). After scoring these headlines (they had to take at least 20 s for this task), they were forwarded to the manipulation check module.

At the end of the survey, respondents were asked to answer two questions as a manipulation check. First, to test whether respondents correctly identified that the media literacy intervention was designed to identify fake news or recognise trustworthy information, respondents were asked what the article was about. The answer categories were as follows: (1) how to identify fake news; (2) how to recognise trustworthy information and (3) a Rembrandt painting. Chi-square tests showed that the intervention was correctly recognised by participants (Chi square = 545.59, $df = 4$, $p < 0.001$). Second, to determine if respondents recognised whether the intervention was more generic or specifically about the topic, they answered the question 'Was the article specifically about news about the Ukraine war or news in general?' Chi-square test indicated a successful manipulation of the specificity of the intervention (Chi square = 121.11, $df = 2$, $p < .001$).

After completing all questions, participants were carefully debriefed, which included showing them the fact-checks of all the headlines that they scored, and a general message emphasising that they were exposed to a number of false and true statements for the purposes of measuring the effect of a warning message on people's ability to detect false information.

Table 2. OLS-regression models predicting truth ratings after exposure to interventions.

	Dependent variable					
	Truth rating misinformation		Truth rating factually accurate information		Composite truth rating	
	(Model 1)	(Model 2)	(Model 3)	(Model 4)	(Model 5)	(Model 6)
Condition 1: Trust NML (general)	−0.36 (0.28)	−0.35 (0.61)	0.07 (0.25)	−0.25 (0.55)	0.42 (0.33)	0.08 (0.70)
Condition 2: Trust NML (specific)	0.04 (0.28)	−0.06 (0.63)	−0.03 (0.25)	−0.59 (0.56)	−0.07 (0.33)	−0.58 (0.72)
Condition 3: Warning NML (general)	−0.13 (0.28)	−0.92 (0.64)	−0.08 (0.25)	−1.26** (0.57)	0.05 (0.33)	−0.32 (0.73)
Condition 4: Warning NML (specific)	−0.24 (0.28)	−1.27** (0.62)	−0.19 (0.25)	−0.66 (0.55)	0.05 (0.33)	0.54 (0.71)
Media trust		0.13 (0.11)		0.07 (0.10)		−0.03 (0.13)
Political ideology (ranging from center to right)		0.28**** (0.03)		−0.06** (0.03)		−0.34**** (0.04)
Political interest		−0.15*** (0.05)		0.23**** (0.04)		0.40**** (0.06)
Condition 1: Trust NML (general) * media trust		−0.01 (0.15)		0.09 (0.14)		0.11 (0.18)
Condition 2: Trust NML (specific) * media trust		0.02 (0.16)		0.15 (0.14)		0.14 (0.18)
Condition 3: Warning NML (general) * media trust		0.21 (0.16)		0.30** (0.14)		0.09 (0.18)
Condition 4: Warning NML (specific) * media trust		0.29* (0.16)		0.14 (0.14)		−0.13 (0.18)
Constant	−0.57*** (0.20)	−1.12* (0.58)	1.33**** (0.18)	0.33 (0.44)	1.89**** (0.23)	2.07**** (0.57)
Observations	1105	1105	1105	1105	1105	1105
R ²	0.002	0.09	0.001	0.07	0.002	0.11
Adjusted R ²	−0.001	0.08	−0.003	0.06	−0.001	0.10
Residual Std. Error	3.00 (df = 1100)	2.88 (df = 1093)	2.64 (df = 1100)	2.56 (df = 1093)	3.47 (df = 1100)	3.30 (df = 1093)
F-Statistic	0.66 (df = 4; 1100)	8.65**** (df = 12; 1093)	0.29 (df = 4; 1100)	7.22**** (df = 11; 1093)	0.69 (df = 4; 1100)	11.90**** (df = 11; 1093)

* $p < .1$; ** $p < .05$; *** $p < .01$; **** $p < .001$.

3. Results

We expected that, compared to the absence of an intervention, exposure to a media literacy intervention would lower the credibility of both factually accurate headlines (H1a) and headlines of misinformation articles (H1b). To test the effectiveness of exposure to the different interventions designed for this experiment, we ran a series of OLS regression models (see Table 2). This table summarises the outcomes of the regression analyses respectively predicting the truth rating of misinformation (Model 1), factually accurate information (Model 3) and the composite score reflecting the truth rating of both factually accurate information and misinformation (Model 5). As can be seen in the table, none of the predictors are significant. Hence, *ceteris paribus*, exposure to the different interventions did not influence the truth rating of factually accurate information or misinformation. Against this backdrop, our findings do not offer any support for H1a, or H1b. In addition, as there is no effect of any of the conditions, RQ₁ about the overall effects of interventions, or H2 about the differential effects of specific versus general warnings, are not supported by the data.

For H3, we split up the analyses for the truth rating of specific and general misinformation (Table 3) and factually accurate (Table 4) statements. Of the eight misinformation statements, four were about the war in Ukraine and four were about more general topics. The same logic applies to the eight factually accurate statements. Models 1 and 3 in Table 3 show no significant differences across experimental groups comparing truth ratings of specific versus general misinformation statements. Similarly, Models 1 and 3 in Table 4 show no difference across experimental groups comparing truth ratings of specific versus general factually accurate statements. These findings do not support H3a-b.

3.1. Additional analyses

Although we pre-registered the hypotheses for this study, our pre-analysis plan did not specify any moderation effects. We also did not pre-register the use of additional controls in our models. Given the lack of direct findings, we explored whether suppression by measured variables caused the main findings to be non-substantial. Hence, it could be the case that the media literacy interventions were only effective for

Table 3. OLS-regression models predicting truth of specific and general misinformation.

	Dependent variable			
	Truth rating misinformation specifically about Ukraine war		Truth rating general misinformation	
	(Model 1)	(Model 2)	(Model 3)	(Model 4)
Condition 1: Trust NML (general)	−0.11 (0.17)	−0.20 (0.38)	−0.25 (0.17)	−0.15 (0.36)
Condition 2: Trust NML (specific)	0.14 (0.17)	−0.04 (0.39)	−0.10 (0.17)	−0.02 (0.37)
Condition 3: Warning NML (general)	−0.06 (0.17)	−0.19 (0.40)	−0.08 (0.17)	−0.72* (0.38)
Condition 4: Warning NML (specific)	−0.06 (0.17)	−0.75* (0.39)	−0.18 (0.17)	−0.52 (0.37)
Media trust		0.18** (0.07)		−0.05 (0.07)
Political ideology (ranging from left to right)		0.08**** (0.02)		0.19**** (0.02)
Political interest		−0.11**** (0.03)		−0.04 (0.03)
Condition 1: Trust NML (general) * media trust		0.02 (0.10)		−0.03 (0.09)
Condition 2: Trust NML (specific) * media trust		0.05 (0.10)		−0.02 (0.09)
Condition 3: Warning NML (general) * media trust		0.03 (0.10)		0.17* (0.09)
Condition 4: Warning NML (specific) * media trust		0.19* (0.10)		0.10 (0.09)
Constant	−0.54**** (0.12)	−0.75** (0.36)	−0.03 (0.12)	−0.37 (0.34)
Observations	1105	1105	1105	1105
R^2	0.002	0.06	0.002	0.10
Adjusted R^2	−0.001	0.05	−0.001	0.09
Residual Std. Error	1.84 (df = 1100)	1.78 (df = 1093)	1.78 (df = 1100)	1.69 (df = 1093)
F-Statistic	0.59 (df = 4; 1100)	6.18**** (df = 12; 1032)	0.65 (df = 4; 1100)	10.26**** (df = 12; 1093)

* $p < .1$; ** $p < .05$; *** $p < .01$; **** $p < .001$.

some participants, whereas others rejected or even counter-argued them. For this reason, we ran additional analyses for which we added several control variables (political ideology and political interest) and a moderating variable to the model (media trust). We specifically expect that media trust can act as an important factor that influences participants' rating of factually accurate and false information. Hence, participants with more trust in the media may be more likely to follow the suggestions of the media literacy interventions, whereas distrusting participants may disregard their suggestions. Participants that distrust the media may perceive that established sources of information – including sources communicating media literacy interventions – are not able to fulfil their expectation to inform in an accurate and honest way. This may lead them to not consider the warnings presented in the interventions we exposed participants to.

As can be seen in Models 2, 4 and 6 of Table 2, when taking these factors into account, exposure to media literacy interventions can influence respondents' truth rating of misinformation and factually accurate information. For the assessment of misinformation (Model 2), we see that a specific warning for misinformation about the war in Ukraine can be successful in

decreasing beliefs in misinformation ($b = -1.27$, $p < .05$). Respondents exposed to this intervention were less likely to rate misinformation as true, which supports H1a. When looking at the truth rating of factually accurate information, however, a different picture arises. More specifically, factually accurate information can be rated as less accurate when participants are exposed to a general warning for misinformation ($b = -1.26$, $p < .05$, Model 4). Hence, participants exposed to this intervention were less likely to rate factually accurate information as true, potentially because they became too critical or afraid of being misinformed. This supports H1b. When looking at the composite score of truth discernment (Model 6), we see that these effects disappear. Hence, in line with RQ₁, it can be argued that the intended effect on accuracy for the truth rating of misinformation and the unintended effect of the media literacy message on lowering trust in factually accurate information result in a non-existent net effect on truth discernment.

Next, when looking at the interaction effect between exposure to a media literacy intervention and media trust, our findings indicate that the specific media literacy warning can backfire in identifying misinformation for those who generally tend to trust media more,

as this warning increased their truth ratings of misinformation statements ($b = .29$, $p < .10$; Model 2, Table 2). Yet, only for those participants that trust the news media, can a general warning for misinformation be successful in increasing their truth ratings of factually accurate information ($b = .30$, $p < .05$; Model 4, Table 1). Thus, the way in which the intervention is presented seems to matter. Specifically, general rather than issue-specific warnings about misinformation are more effective for participants with higher levels of trust in the media. Although this does not offer direct support for H3, issue-specific warning messages are more effective among those who hold lower levels of media trust while general warnings are more effective when targeted at those with higher media trust levels.

If we look at Tables 3 and 4, where the dependent variable of truth rating is split up to present issue specific or general misinformation and factually correct statements, a similar picture arises. When taking the moderating and control variables into account, those who were exposed to an issue-specific warning would only rate misinformation as less credible ($b = -.75$, $p < .10$; Table 3, Model 2) and the general warning would only decrease credibility ratings of general misinformation ($b = -.72$, $p < .10$; Table 3, Model 4). For credibility ratings of factually accurate information, only a general warning was found to backfire and decrease credibility ratings for the general statement ($b = -1.03$, $p < .001$; Table 4, Model 4).

4. Discussion

Although pre-warnings for misinformation and media literacy messages may be effective in correcting misperceptions by enhancing resilience to misinformation (e.g. Guess et al. 2020; Roozenbeek and van der Linden 2019), they may also have a negative impact on trust in accurate information (e.g. Van der Meer, Hameleers, and Ohme 2023). More specifically, by priming suspicion about incoming deception, pre-bunking interventions may motivate people to deviate from the truth-default, even when exposed to factually accurate information. To test whether such side-effects occur in the context of media literacy interventions, we report the findings of an experiment in which we exposed people to different types of media literacy interventions after which they rated the veracity of both factually accurate and false headlines on different issues.

Our main findings indicate that exposure to a media literacy message that warns about misinformation generally has no straightforward effects on the truth rating of misinformation or factually accurate information. Given that these findings are not in line with prior

studies on the effectiveness of media literacy interventions in warning people about misinformation (e.g. Guess et al. 2020; Hameleers 2022; Vraga, Bode, and Tully 2022), this study echoes conclusions regarding the complexity of stimulating certain behaviours with news media literacy interventions (Van der Meer, Hameleers, and Ohme 2023). However, it can be noted that previous studies found only small effects of exposure to warnings on the credibility rating of misinformation, even when participants were exposed to a very clear misinformation article after reading the warning message (Hameleers 2022).

However, when we consider pre-existing levels of media trust, exposure to issue-specific warnings generally lower the credibility of misinformation, whereas general warnings can lower the truth rating of factually accurate information. This finding is in line with recent literature indicating that pre-bunking interventions for misinformation may backfire under some conditions by lowering trust in factually accurate information (e.g. Acerbi, Altay, and Mercier 2022; Modirrousta-Galian and Higham 2022; Van der Meer, Hameleers, and Ohme 2023). Hence, for people that tend to trust the media, specific warnings about misinformation may trigger suspicion that is also applied to the truth rating of accurate news. In line with the truth-default theory (Levine 2014), the media literacy message may function as a prime that activates suspicion related to potential deception, which makes trusting news users overtly cynical to the headlines they are rating. This argues for tailoring news media literacy interventions according to people's pre-existing level of media trust, as those with lower trust can have opposite reactions to those already trusting media, where interventions are sometimes successful and other times backfire.

Our findings contribute to the media literacy and misinformation literature by showing that making interventions that pre-bunk misinformation effective across the board is a rather complex endeavour. We experimented with different formats of warning about misinformation, aiming to overcome the negative effect of media literacy messages on the truth rating of accurate information (Van der Meer, Hameleers, and Ohme 2023). Yet, our findings indicate that messages that warn about misinformation whilst relativising trust in factually accurate information are not more effective in enhancing the accuracy of discernment between misinformation and factually accurate news, indicating the challenges in increasing general trust in news and improving deception detection with a single media literacy intervention. However, the warning's level of specificity can make a difference. A very general warning about misinformation may reduce trust in

Table 4. OLS-regression models predicting truth of specific and general factually accurate information.

	Dependent variable:			
	Truth rating factually accurate information specifically about war in Ukraine		Truth rating general factually accurate information	
	(Model 1)	(Model 2)	(Model 3)	(Model 4)
Condition 1: Trust NML (general)	0.10 (0.16)	0.08 (0.37)	−0.03 (0.16)	−0.33 (0.34)
Condition 2: Trust NML (specific)	−0.05 (0.17)	−0.26 (0.38)	0.01 (0.16)	−0.32 (0.35)
Condition 3: Warning NML (general)	−0.07 (0.17)	−0.23 (0.38)	−0.01 (0.16)	−1.03*** (0.36)
Condition 4: Warning NML (specific)	−0.08 (0.17)	−0.16 (0.37)	−0.11 (0.16)	−0.50 (0.35)
Media trust		−0.02 (0.07)		0.09 (0.06)
Political ideology (ranging from left to right)		−0.06**** (0.02)		0.003 (0.02)
Political interest		0.15**** (0.03)		0.08*** (0.03)
Condition 1: Trust NML (general) * media trust		0.01 (0.09)		0.08 (0.09)
Condition 2: Trust NML (specific) * media trust		0.06 (0.09)		0.09 (0.09)
Condition 3: Warning NML (general) * media trust		0.04 (0.09)		0.26*** (0.09)
Condition 4: Warning NML (specific) * media trust		0.03 (0.09)		0.11 (0.09)
Constant	1.45**** (0.12)	1.16**** (0.30)	−0.13 (0.11)	−0.83*** (0.28)
Observations	1105	1105	1105	1105
R^2	0.001	0.03	0.001	0.07
Adjusted R^2	−0.002	0.02	−0.003	0.06
Residual Std. Error	1.74 (df = 1100)	1.72 (df = 1093)	1.67 (df = 1100)	1.61 (df = 1093)
F-Statistic	0.39 (df = 4; 1100)	3.50**** (df = 11; 1093)	0.19 (df = 4; 1100)	7.87**** (df = 11; 1093)

*** $p < .01$; **** $p < .001$.

factually accurate information, whereas this is not the case for an issue-specific warning.

These findings have practical implications. Although our findings are less optimistic about the potential of pre-bunking interventions compared to earlier research (e.g. Guess et al. 2020), we suggest media literacy educators and policymakers to work with issue-specific warnings as much as possible, and tailor warning messages about misinformation to the pre-existing levels of trust of the intended audience. Thus, as general warnings about misinformation without specifying the issue the warning relates to may negatively impact the truth rating of accurate information, we suggest developing different interventions for different issues. Concretely, interventions may be tailored to deal with issues that are likely to be surrounded by misinformation, such as immigration, COVID-19, or the Russian war in Ukraine. For these salient issues, specific warnings that relate to the nature of misinformation on the respective topic may be developed. For example, misinformation on immigration may often refer to numbers about increasing crime rates, violent acts, or the dangers of immigration. A tailored issue-specific warning message may flag some of these suspicious narratives and

connections to stimulate critical thinking among recipients. Yet, the appropriate strategy of media literacy interventions may depend on the trust audiences have in the media, which makes the tailoring of messages crucial.

Our findings come with a number of limitations. First, we used headlines of misinformation instead of actual misinformation narratives. Although this set-up allowed us to use a larger number of fact-checked false statements, it may be rather difficult to rate the truth value of information based on the headline only. Hence, people may arrive at better-informed judgments if they can evaluate statements in the context of an entire message. This is reflected in the many uncertain responses of participants that had to rate whether the headlines were true or false. We suggest future research to extend the truth discernment task to short summaries or first sentences of news articles. In addition, it may be worthwhile to vary the difficulty of the statements: Some headlines may be easier to detect as true or false, whereas others are formulated in a less verifiable manner. Second, we only exposed people to one short media literacy message, and directly forwarded them to the task of rating headlines. In reality, enhancing media literacy is part of a more

encompassing educational package with different messages, modules and assignments. Media literacy goes beyond the ability to discern whether a message is true or false, as it also involves skills and competencies needed to navigate the (digital) media landscape, and critically consider how messages are produced, what biases may be involved in the selection and production of information, and how recipients' own framework plays a role in the interpretation of information (e.g. Livingstone 2014). The message used in this study may not comprehensively deal with the needed competencies and skills, and may demonstrate a bias towards the specific context of the politicised statements. We suggest future research to experiment with more elaborate media literacy packages.

Third, we only measured people's truth ratings on the short term, which overlooks the longer-term effects of exposure to media literacy messages and misinformation. Future research should explore the role of repeated exposure to interventions, as well as a delay between a media literacy intervention and the actual rating of statements.

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